

# **FAILSAFE**

Early Student Failure Detection System

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May 18, 2026

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# Chapter 1

## The Motivation Behind FAILSAFE

Educational institutions often detect academic failure only after end-semester results are released. By the time faculty identify struggling students, opportunities for meaningful intervention have already diminished. This creates a fundamentally reactive educational environment where institutional support arrives after academic decline has already occurred.

The central motivation behind FAILSAFE was therefore not merely prediction accuracy, but proactive intervention.

The project attempts to answer the following question:

*“Can machine learning identify academically vulnerable students before final academic failure occurs?”*

Traditional academic evaluation systems rely heavily on final grades, while many hidden indicators of academic distress remain underutilized:

- attendance patterns,
- study habits,
- alcohol consumption,
- family background,
- travel time,
- social behavior,
- prior academic failures.

FAILSAFE was conceptualized as an explainable educational intelligence system capable of:

- early student risk prediction,
- interpretable machine learning analysis,

- faculty-facing dashboards,
- and automated intervention recommendation.

Rather than functioning as a black-box classifier, the system was designed to provide actionable reasoning behind every prediction.

## Chapter 2

# Constructing the Dataset Universe

The project utilized the UCI Student Performance Dataset containing 395 student records and 33 features.

The dataset contains demographic, behavioral and academic indicators such as:

- attendance,
- study time,
- family relationships,
- internet access,
- alcohol consumption,
- health,
- previous academic failures,
- free time,
- social activity.

The original dataset contains three grade variables:

- $G1$ ,
- $G2$ ,
- $G3$ .

The final grade  $G3$  was used to construct the target variable:

$$\text{at risk} = \begin{cases} 1 & \text{if } G3 < 10 \\ 0 & \text{otherwise} \end{cases}$$

where:

- 1 represents a student at academic risk,
- 0 represents a student not at academic risk.

To prevent target leakage, the variables:

$$G1, G2, G3$$

were removed during training. This ensured that the model learned behavioral and environmental predictors rather than directly learning grading correlations.

Categorical variables such as:

- school,
- guardian,
- address,
- internet,
- romantic

were transformed numerically using Label Encoding:

$$x_i \rightarrow \{0, 1, 2, \dots, n\}$$

allowing tree-based algorithms to process categorical information efficiently.

# Chapter 3

## Building Predictive Intelligence

The predictive engine of FAILSAFE was implemented using Extreme Gradient Boosting (XGBoost).

XGBoost is an ensemble learning algorithm that combines multiple weak decision trees into a strong predictive framework.

The final prediction is represented mathematically as:

$$\hat{y}_i = \sum_{k=1}^K f_k(x_i)$$

where:

- $f_k$  represents the  $k^{th}$  decision tree,
- $K$  is the total number of boosting rounds,
- $x_i$  is the feature vector for student  $i$ .

Unlike traditional decision trees, boosting algorithms iteratively minimize prediction residuals.

At iteration  $t$ :

$$L^{(t)} = \sum_i l\left(y_i, \hat{y}_i^{(t-1)} + f_t(x_i)\right) + \Omega(f_t)$$

where:

- $l$  represents the classification loss function,
- $\Omega$  represents the regularization term controlling model complexity.

Regularization prevents overfitting by penalizing excessively complex tree structures. The model was trained using:

- 100 estimators,

- maximum tree depth of 4,
- learning rate of 0.1.

XGBoost was selected because:

- it performs exceptionally well on structured tabular data,
- captures non-linear feature interactions,
- handles heterogeneous feature spaces,
- and provides strong predictive performance with limited datasets.

The training pipeline included:

- train-test splitting,
- stratified sampling,
- probability-based prediction,
- ROC-AUC evaluation.



# Chapter 4

## Explainability Through SHAP

A major problem with many machine learning systems is interpretability.

Educational institutions cannot rely on purely opaque predictions for sensitive academic intervention decisions.

To overcome this limitation, FAILSAFE integrated SHAP:

SHapley Additive exPlanations

SHAP originates from cooperative game theory.

The central idea is that each feature contributes a measurable amount toward the final prediction.

For a prediction function  $f(x)$ :

$$f(x) = \phi_0 + \sum_{i=1}^M \phi_i$$

where:

- $\phi_0$  represents the baseline prediction,
- $\phi_i$  represents the contribution of feature  $i$ ,
- $M$  is the number of features.

SHAP therefore decomposes every prediction into additive feature contributions.

The system generated:

- global feature importance,
- per-student explanations,
- directional feature influence.

The most influential features identified included:

- failures,

- absences,
- goout,
- freetime,
- family relationships.

For instance:

- high absences positively increased risk probability,
- higher study time negatively reduced risk probability.

This transformed FAILSAFE from a black-box classifier into an interpretable educational analytics system.

# Chapter 5

## Intervention Recommendation Engine

Prediction alone is insufficient in educational systems.

The project therefore introduced a rule-based intervention engine layered over SHAP outputs.

The top contributing SHAP features were mapped to actionable faculty recommendations.

Examples included:

- High absences → attendance counselling,
- Multiple failures → peer tutoring,
- High alcohol indicators → wellness counselling,
- Low study time → structured study plans,
- Poor family relations → student support services.

The intervention engine transformed the pipeline from:

Prediction System

into:

Decision Support Framework

capable of generating actionable educational responses.

# Chapter 6

## System Architecture and Deployment

The FAILSAFE system architecture can be summarized as:

Faculty CSV Upload → Preprocessing and Encoding → XGBoost Prediction → SHAP  
Explainability → Intervention Engine → Risk Dashboard

The frontend was developed using Streamlit because of:

- native Python integration,
- rapid dashboard construction,
- lightweight deployment capabilities,
- real-time visualization support.

Two major operational modules were implemented:

### 1. Bulk Upload Dashboard

Faculty upload CSV files containing student data and receive ranked risk predictions for all students.

### 2. Individual Risk Dashboard

Faculty can inspect individual students, visualize SHAP explanations and view personalized intervention plans.

The application was deployed using:

GitHub → HuggingFace Spaces → Public Streamlit Deployment

HuggingFace Spaces was selected instead of GitHub Pages because GitHub Pages only supports static web hosting.

FAILSAFE required:

- Python execution,

- XGBoost inference,
- SHAP computation,
- Streamlit runtime,
- real-time prediction pipelines.

HuggingFace Spaces provided:

- cloud-based ML hosting,
- Streamlit support,
- simplified deployment,
- public accessibility.

Live Demo:

<https://huggingface.co/spaces/thisiskartik05/FailSafev1>

GitHub Repository:

<https://github.com/thisiskartik05/FailSafe2026>

# Chapter 7

## Experimental Results

The trained model achieved the following performance:

| Metric              | Value  |
|---------------------|--------|
| Accuracy            | 0.72   |
| AUC Score           | 0.7097 |
| Weighted F1 Score   | 0.70   |
| Precision (At Risk) | 0.62   |
| Recall (At Risk)    | 0.38   |
| F1 Score (At Risk)  | 0.48   |

The confusion-sensitive metrics indicate that the system successfully identified academically vulnerable students with reasonable predictive reliability despite the relatively small dataset size.

The ROC-AUC score of:

0.7097

demonstrates that the classifier performs substantially better than random prediction. Additionally, SHAP analysis revealed that:

- prior failures,
- attendance,
- social behavior,
- and free-time patterns

were among the strongest predictors of academic vulnerability.

# Chapter 8

## Conclusion

FAILSAFE demonstrates how explainable machine learning can support proactive educational intervention systems.

The project evolved beyond a simple predictive classifier into an interpretable educational analytics framework integrating:

- XGBoost classification,
- SHAP explainability,
- intervention recommendation systems,
- faculty-facing dashboards,
- cloud deployment engineering.

The combination of:

- predictive intelligence,
- mathematical explainability,
- and actionable intervention

creates a practical system capable of assisting faculty in identifying academically vulnerable students before academic decline becomes irreversible.

Future improvements may include:

- longitudinal academic tracking,
- deep learning temporal models,
- institution-specific retraining,
- reinforcement-based intervention optimization,
- real-time academic monitoring.

Ultimately, FAILSAFE attempts to shift educational systems from reactive grading toward proactive student support.